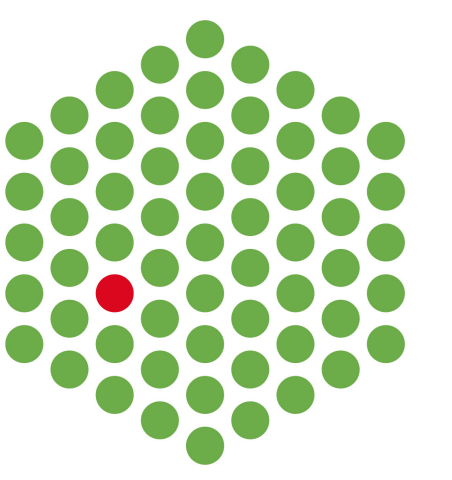


Chemical probes in the scientific literature: expanding and validating target-disease evidence



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Chemical probes in the scientific literature

Chemical probes are **potent and selective small molecules** that bind specifically to a **protein target of interest**. They serve as indispensable tools in drug discovery as they offer the means to investigate the functional role of a protein in a cellular context, helping to establish a **causal link between the target and a disease (T-D link)**. This information is crucial in prioritizing targets for further drug development efforts. Thousands of these experiments are annually reported in the literature, but the data is buried in unstructured text. To explore such data, we have developed an NLP pipeline that extracts chemical probe-Target-Disease links from articles.

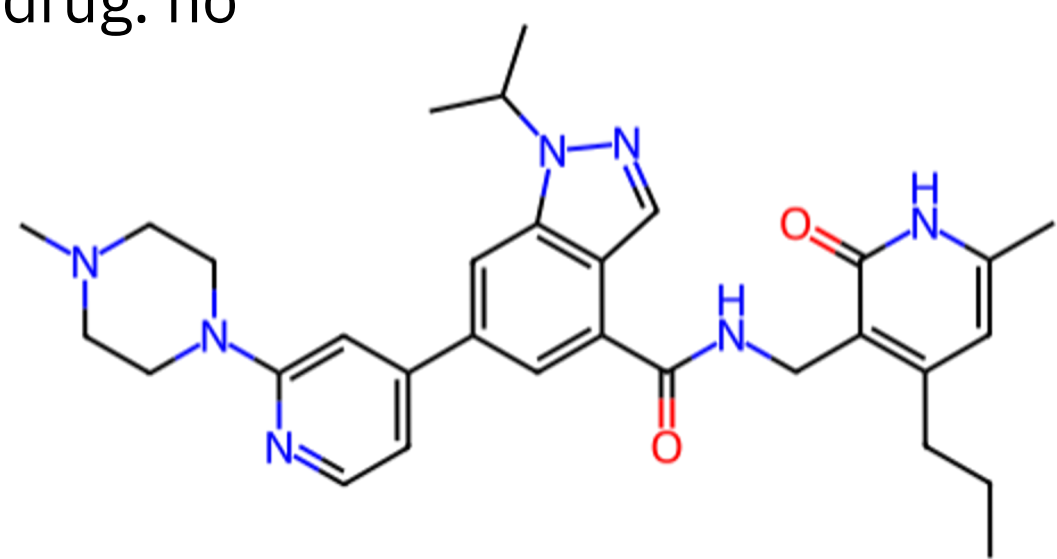
GSK343

Defined target: EZH2

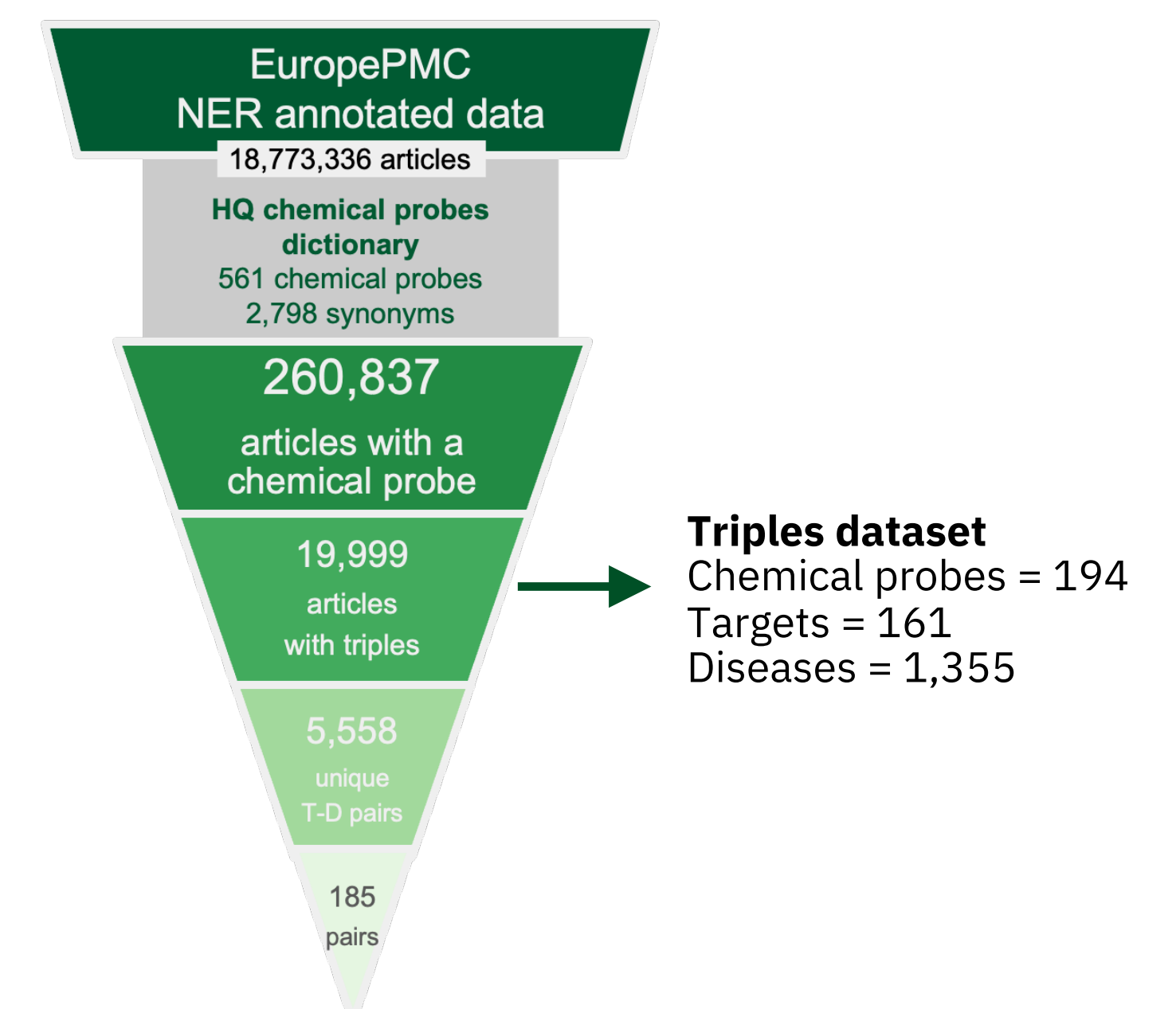
Mode of action: SAM competitive

Recommended concentration: 100 nM - 3 µM

Is a drug: no

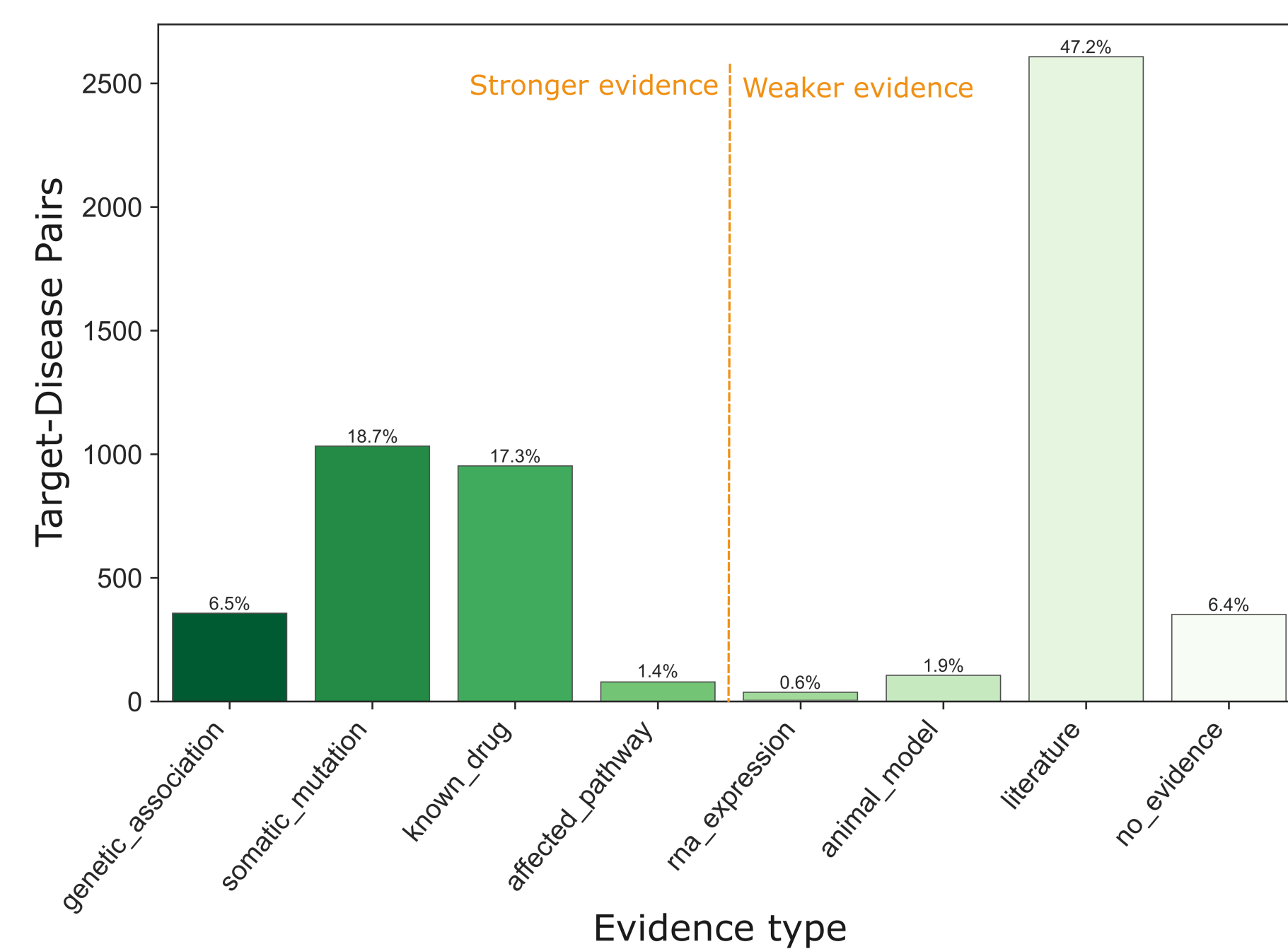


Chemical probes provide the essential functional and pharmacological validation required to establish the link between a target and a disease



Chemical probes data reinforces the existing target-disease evidence

Q: Have chemical probes data (Target-Disease links) been reported in any other type of evidence?

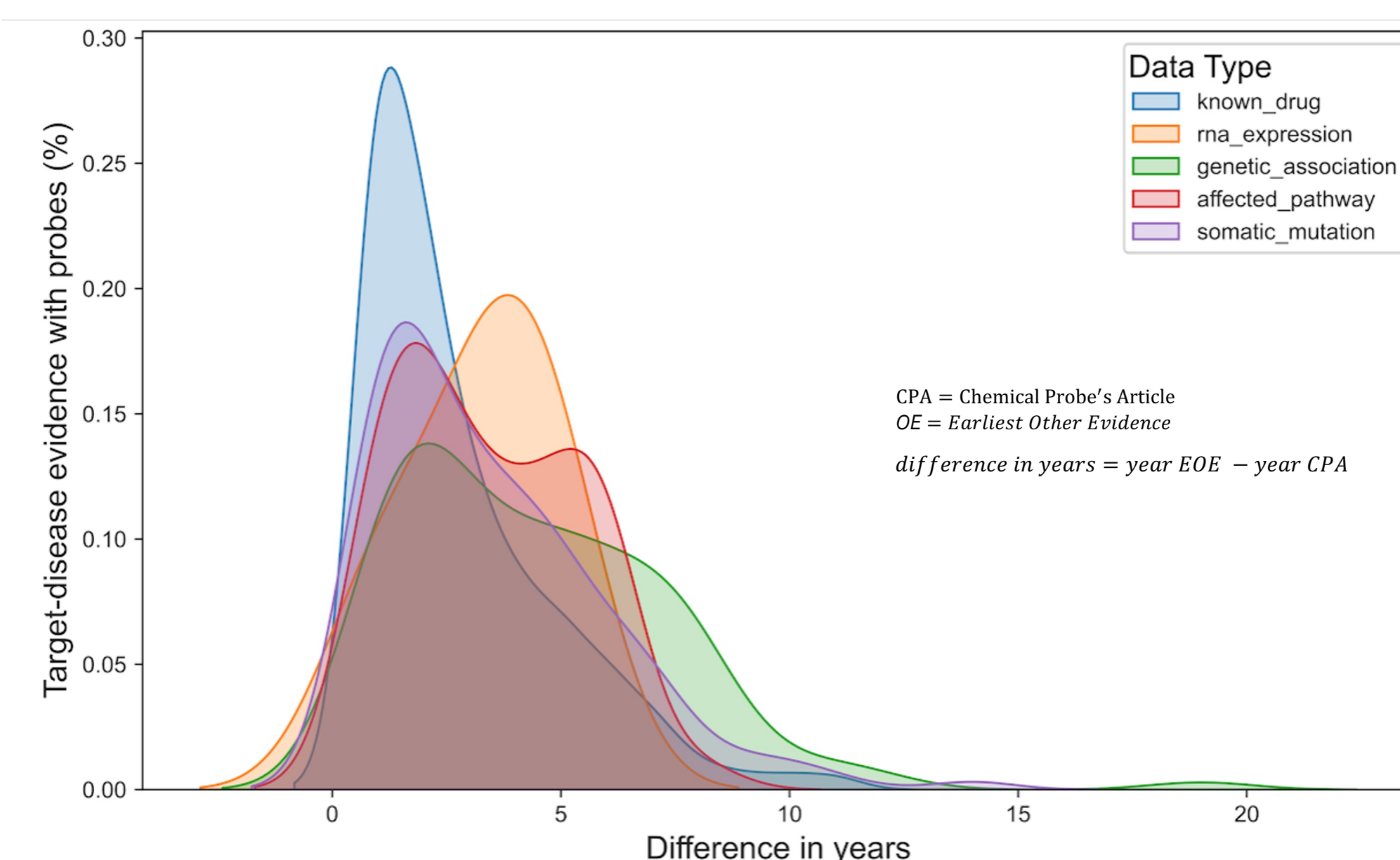


5,558 Target-Disease pairs found in chemical probes literature:

- 43.8% had strong evidence
- 2.5% had weaker evidence
- 47.4% had literature evidence (pure co-occurrence without chemical probes' data)
- 6.4% no previous evidence

R: >50% of cases is new evidence, or it elevated existing evidence with functional validation

Q: If other evidence was found for a target-disease link, was it reported before or after the chemical probes article?



~2000 Target-Disease pairs with other non-literature evidence

185 Target-Disease pairs with chemical probe's early evidence (the chemical probe evidence was before other evidence)

R: In many cases, chemical probe's evidence anticipates structured database annotations from up to 20 years

Examples

Target	Disease	Chemical probe	Article (PMDID)	Publication date	Other evidence year	Time lag
CCKBR	duodenal ulcer	L-365260	9892884	01/01/1999	2018	19
MTOR	type 2 diabetes mellitus	Rapamycin	18174523	03/01/2008	2020	12
EDNRB	diabetes mellitus	A-192621	18287215	20/02/2008	2019	11
JAK2	splenomegaly	Ruxolitinib	19468275	01/03/2009	2020	11
KCNJ11	neonatal diabetes mellitus	Glyburide	15531505	01/11/2004	2013	9

Truncated table

Article sentence
"mTOR inhibition by rapamycin prevents beta-cell adaptation to hyperglycemia and exacerbates the metabolic state in type 2 diabetes."
Other evidence
 Open Targets Genetic Association

Chemical probes as early predictors of novel therapeutic opportunities

Research area	Target	Disease/Phenotype	Chemical probes	Articles
Therapeutic Repurposing in non-oncological indications	CDK4	Acute & Chronic Kidney Disease	2	8
	PARP1	Parkinson's and Multiple Sclerosis	2	4
	NR1H3 (LXR)	NAFLD and Insulin Resistance	2	4
Precision approaches in rare and refractory autoimmunity	JAK1 / JAK2	COPA Syndrome and STING disease	2	5
	JAK1 / JAK2	Still's disease, IgG4-related disease and Antisynthetase syndrome	2	13
	JAK1	Alopecia totalis & universalis	2	3
Targeted strategies in defined oncogenic contexts	MDM2	Rhabdoid tumours and Brain cancer	3	5
	EZH2	Glioma and Retinoblastoma	2	2
	SOS1	AML	2	2
	IGF1R	Malignant pleural mesothelioma	2	2

Strict criteria to identify therapeutic opportunities

- **Robust:** Target-Disease link reported with ≥2 distinct probes
- **Recent:** First probe's publication <5 years ago
- **Understudied:** Link with ≤20 supporting articles
- **Novel:** No approved drugs or active clinical programs.

Examples 135 high-confidence Target-Disease pairs

Conclusion

The chemical probes literature unlocks hidden academic experiments, providing critical early validation of Target-Disease links, a predictive advantage over structured databases in key cases, and robust novel therapeutic opportunities.

Preprint here!

